Maximizing Solar Panel Output for a Fixed Area

In this project, you will use MATLAB to formulate and solve an optimization problem: Given a fixed area, determine the optimal tilt angle and aspect ratio of a solar panel to maximize the total energy output.

Use this Live Script as a template to get started on your project solution. You may include helper functions at the bottom of this document or as separate files. For a tutorial on how to use Live Scripts, check out this video.

Suggested Tasks

1. Define the objective function in MATLAB (the energy function).
2. Use a problem-based optimization workflow to find the values of θ and r that maximize the energy output. Constrain the values: $0 \leq \theta \leq 90$ and $0.5 \leq r \leq 4$
3. Plot the objective function using `meshgrid` and `surf` to visualize E(θ, r) as a 3D surface plot. Then, print the optimal angle, ratio, and corresponding energy output.

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Break Down the Problem

In the space below, define the problem statement:

Solar panel efficiency is dependent upon numerous factors. Using the energy function, two salient variables are considered: tilt angle and aspect ratio. Thus, to produce the highest amount of energy from our solar panel, the optimal tilt angle and aspect ratio respectively must be found.

List the requirements, constraints, and success criteria for the project:

To find the optimal tilt angle and aspect ratio for the solar panel, an optimization function for both values is needed. We opted for MATLAB's Optimization Toolbox to find the values for which our energy output is the greatest. Similarly, the area of our solar panel is two squared meters, our angle is between zero and ninety degrees, and the solar panel aspect ratio is between 0.5 and 4. The solution to this project is the verified optimal tilt angle and aspect ratio for the maximum solar panel output.

What is your proposed solution?

By using optimization functions included in MATLAB, given the constraints, the optimal tilt angle and surface area values can be found. Our solution can then be compared with a three-dimensional surface plot of different tilt angles and surface areas with their associated energy output.

List the steps you will need to take to achieve your proposed solution:

Using MATLAB, we will first define our energy function as well as its associated subfunctions, area (A), sun intensity (sunIntensity), the efficiency function based on angle (nu), and the shape efficiency factor based on the aspect ratio (f). Next, we will model the solution using MATLAB's Optimization Toolbox functions, maximizing energy output given our tilt angle and aspect ratio variables. After finding the greatest energy output, we will then plot the energy output against tilt angle and aspect ratio in a three-dimensional graph and compare our solution.

What challenges do you face in the process of achieving the solution?

Some problems faced during this project were communication among the team. We found each other at various stages in understanding the technology for the completion of this project. Especially with version control via GitHub and using MATLAB. Similarly, we faced issues with task delegation and version management throughout the project. Through proper leadership, we were able to consolidate tasks and find accurate results.

Task 1: Define the Objective Function

The energy function is the objective function for this optimization problem. It calculates how much energy a solar panel produces based on its tilt angle (theta) and aspect ratio (r). Check the Project Description for the energy formula and parameter definitions and use the image below as a guide:

$$\boxed{A = 2m^2}$$

$$\boxed{sunIntensity(\theta) = 1000 * \cos(\theta - 45)}$$

$$E(\theta, r) = A \cdot \eta(\theta) \cdot sunIntensity(\theta) \cdot f(r)$$

$$\boxed{f(r) = \exp(-0.1 * (r-1)^2)}$$

$$\boxed{\eta(\theta) = \cos(\theta - 30)}$$

```
% Insert your code in the spaces provided (or make use of helper functions and
indicate where
% they can be found). Ensure your code is well-documented.

% Your energy function should output energy units and take as inputs tilt
% angle (theta) and aspect ratio (r). Using the intermediary steps can help
% simplify how you define E.

% Step 1: Define A (panel area in m^2)

A = 2;

% Step 2: Define nu (efficiency function based on tilt angle)

nu = @(theta) cosd(theta - 30);
```

```
% Step 3: Define sunIntensity (sunlight variation with tilt)

sunIntensity = @(theta) 1000 .* cosd(theta - 45);

% Step 4: Define fr (shape efficiency factor based on aspect ratio)

fr = @(r) exp(-0.1 .* (r - 1).^2);

% Step 5: Calculate E (total energy output)

E = @(theta, r) A .* nu(theta) .* sunIntensity(theta) .* fr(r);
```

Task 2: Implement the Optimization Function

Use MATLAB's Optimization Toolbox to find the optimal tilt angle and aspect ratio that MAXIMIZE energy output. This MATLAB doc page will be helpful.

```
% Insert your code in the spaces provided (or make use of helper functions and
indicate where
% they can be found). Ensure your code is well-documented. The intermediary
% steps below take you through a problem-based optimization workflow.

% Step 1. Create the optimization problem. We're telling MATLAB: "We want to
MAXIMIZE something."
% That "something" will be energy.

prob = optimproblem(ObjectiveSense="maximize");

% Step 2. Define the optimization variables we can change in the problem
% These are the values MATLAB will try to optimize:
% - theta (tilt angle), must stay between 0 and 90 degrees
% - r (aspect ratio), must stay between 0.5 and 4

theta = optimvar("theta", LowerBound=0, UpperBound=90);

r = optimvar("r", LowerBound=0.5, UpperBound=4);

% Step 3. Turn our energy function into an optimization problem MATLAB can work
% with.
% This step wraps our energy function so MATLAB can plug in different
% theta and r values while solving the problem.
% note: we're optimizing our energy function, so we should try to convert
% that function into an optimization expression

energyOpt = fcn2optimexpr(E, theta, r);

% Step 4. Tell MATLAB what we want to maximize
% We set our energy formula as the "objective" of the problem
```

```
% (this is what we want the solver to maximize)

prob.Objective = energyOpt;

% Step 5. Give MATLAB a starting point to begin the search
% Optimization needs a guess to start from:
     % Start searching from a tilt angle of ? degrees
     % Start searching from an aspect ratio of ?

start.theta = 45;

start.r = 1;

% Step 6. Now solve the problem!
% Hint: look for a Optimization Toolbox function meant to solve
% expressions. Store the outputs for plotting purposes later on.

[sol, maximizeEnergy] = solve(prob, start);
```

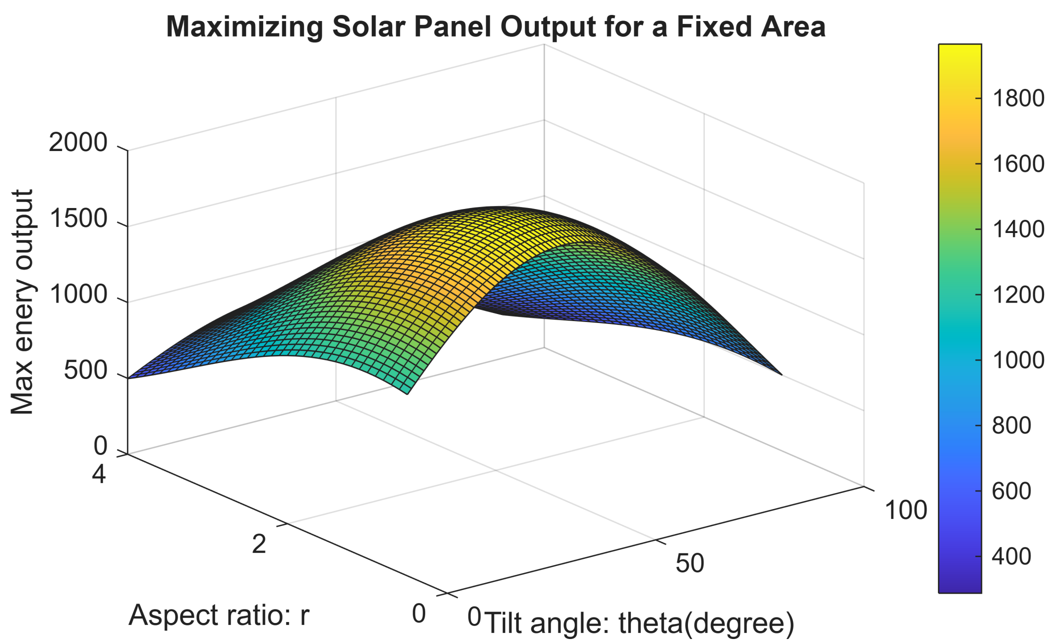
```
Solving problem using fmincon.

Local minimum possible. Constraints satisfied.

fmincon stopped because the size of the current step is less than
the value of the step size tolerance and constraints are
satisfied to within the value of the constraint tolerance.

<stopping criteria details>
```

Task 3: Output the Results

Use the appropriate plotting function to plot E(θ, r) and visualize the energy surface. Print the optimal tilt angle and aspect ratio, and corresponding energy output.

```
% Insert your code in the spaces provided (or make use of helper functions and
indicate where
% they can be found). Ensure your code is well-documented. The intermediary
% steps below will guide you through creating a 3D surface plot.

% Step 1: Make a bunch of points for the x and y axis meshgrid
% The step for the angle is 1, the step for ratio is 0.1
% Hint: keep theta and r within the constraints specified in the problem

thetaAngles = 0: 1: 90;

rValues = 0.5: 0.1: 4;

[thetaGrid, rGrid] = meshgrid(thetaAngles, rValues);
```

```
% Step 2: Calculate energy for every point on the meshgrid using element-wise
operations

EGrid = E(thetaGrid, rGrid);

% Step 3: Draw the 3d graph using the "surf" function

figure;

surf(thetaGrid, rGrid, EGrid);

% Step 4: Label everything on the surf plot (title, axes, etc)

xlabel("Tilt angle: theta(degree)");

ylabel("Aspect ratio: r");

zlabel("Max enery output");

title("Maximizing Solar Panel Output for a Fixed Area");

grid on;

colorbar;
```




```
% Step 5: Print the optimal tilt angle and aspect ratio, and corresponding
% energy output to the command window

fprintf("Optimal Tilt Angle: %.3f degrees\n", sol.theta);
```

```
Optimal Tilt Angle: 37.500 degrees
```

```
fprintf("Optimal Shape (Aspect Ratio): %.3f\n", sol.r);
```

```
Optimal Shape (Aspect Ratio): 1.000
```

```
fprintf("Maximum Energy Output: %.3f Simplified Arbitrary Units\n", maximizeEnergy);
```

```
Maximum Energy Output: 1965.926 Simplified Arbitrary Units
```

Interpretation of Results

Summarize your findings by interpreting their physical/engineering meaning:

The optimal tilt angle was found to be 37.5 degrees and the optimal aspect ratio is 1, indicating a square is the most efficient. This produced a maximum energy output of 1965.926 simplified arbitrary units. With the tilt angle there are two variables contributing to the optimal angle. For tilt efficiency, nu, the best angle is 30 degrees. For the highest sun intensity, it is 45 degrees.

What are some limitations of your work?

This model does not simulate the all real world conditions that could impact the solar panel's efficiency. Some additional factors are location, time of day, season, visibility, temperature, dust, or inverter efficiency when converting DC to AC. The shape efficiency equation fr is a geometric approximation rather than based on physics, environment, or material. The output is in simplified arbitrary units rather than real world units like watts per square meter (W/m^2). This optimization is determined only by the equation, assumptions, and constraints in the model.

What are practical next steps?

Some next steps would be to improve the model by transitioning to real world factors like solar irradiance and weather conditions based upon the solar panel's supposed location. This model could also consider panel azimuth which is the direction the panel is facing, wiring losses, or installation related losses. When the solar cells first meet the solar rays there is a drop in efficiency caused by light induced degradation. Long term effects like long term degradation, dust accumulation, and aging should be included. The model could try different starting points confirming the optimum angle. Also the predicted angle can be evaluated by making a physical prototype for real world values or compare it to existing solar panel data taken from various locations.